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Inventor(s)

MEDINA; Daniel et al.

GROUP-BASED BEAM REPORTING FOR MULTI-RECEIVER OPERATION BASED ON REFERENCE SIGNAL RECEIVED POWER DIFFERENCES

Abstract

An apparatus and method for group-based beam reporting (GBBR) for multi-receiver operation based on reference signal received power (RSRP) differences is provided. The method may include determining, by a first apparatus, a candidate reference signal (RS) group to be provided to at least one second apparatus. The candidate RS group includes at least a first RS and a second RS. The method also includes determining a first difference between a first RSRP of the first RS and a second RSRP of the second RS. The method further includes determining a second difference between a third RSRP of the second RS and a fourth RSRP of the first RS, and performing at least one of excluding the candidate RS group from a group-based beam report based on the first and second differences, or transmitting, to the at least one second apparatus, information indicative of the first and second differences.

Inventors: MEDINA; Daniel (München, DE), BARACCA; Paolo (Munich, DE), CAUDURO DIAS DE PAIVA; Rafael (Aalborg, DK)

Applicant: NOKIA TECHNOLOGIES OY (Espoo, FI)

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Background/Summary

TECHNICAL FIELD

[0001] Some example embodiments may generally relate to mobile or wireless telecommunication systems, such as Long Term Evolution (LTE) or fifth generation (5G) new radio (NR) access technology, or 5G beyond, or sixth generation (6G) access technology or other communications systems. For example, certain example embodiments may relate to group-based beam reporting (GBBR) for multi-receiver (multi-Rx) operation based on reference signal received power (RSRP) differences.

BACKGROUND

[0002] Examples of mobile or wireless telecommunication systems may include the Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (UTRAN), Long Term Evolution (LTE) Evolved UTRAN (E-UTRAN), LTE-Advanced (LTE-A), MulteFire, LTE-A Pro, fifth generation (5G) radio access technology or new radio (NR) access technology and/or sixth generation (6G) radio access technology. Fifth generation (5G) and sixth generation (6G) wireless systems refer to the next generation (NG) of radio systems and network architecture. 5G and 6G network technology is mostly based on new radio (NR) technology, but the 5G/6G (or NG) network can also build on E-UTRAN radio. It is estimated that NR may provide bitrates on the order of 10-20 Gbit/s or higher and may support at least enhanced mobile broadband (eMBB) and ultra-reliable low-latency communication (URLLC) as well as massive machine-type communication (mMTC). NR is expected to deliver extreme broadband and ultra-robust, low-latency connectivity and massive networking to support the Internet of Things (IoT).

SUMMARY

[0003] Various exemplary embodiments may provide an apparatus including at least one processor and at least one memory storing instructions that, when executed by the at least one processor, cause the apparatus at least to determine a candidate reference signal group to be provided to at least one other apparatus. The candidate reference signal group may include at least a first reference signal and a second reference signal. The apparatus may also be caused to determine, based on a first receive beam of the apparatus, a first difference between a first reference signal received power of the first reference signal and a second reference signal received power of the second reference signal, and determine, based on a second receive beam of the apparatus, a second difference between a third reference signal received power of the second reference signal and a fourth reference signal received power of the first reference signal. The method may further include performing at least one of excluding the candidate reference signal group from a group-based beam report based on the first difference and the second difference, or transmitting, to the at least one other apparatus, information indicative of the first difference and the second difference.

[0004] Certain exemplary embodiments may provide a method including determining, by a first apparatus, a candidate reference signal group to be provided to at least one second apparatus. The candidate reference signal group may include at least a first reference signal and a second reference signal. The method may also include determining, by the first apparatus and based on a first receive beam of the first apparatus, a first difference between a first reference signal received power of the first reference signal and a second reference signal received power of the second reference signal, and determining, by the first apparatus and based on a second receive beam of the first apparatus, a second difference between a third reference signal received power of the second reference signal and a fourth reference signal received power of the first reference signal. The method may further include performing at least one of excluding the candidate reference signal group from the group-

based beam report based on the first difference and the second difference, or transmitting, to the at least one second apparatus, information indicative of the first difference and the second difference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] For proper understanding of example embodiments, reference should be made to the accompanying drawings, as follows:

[0006] FIG. 1 illustrates an example of a network configuration for GBBR, according to various exemplary embodiments;

[0007] FIG. 2 illustrates a signal diagram for one or more procedures, according to certain exemplary embodiments;

[0008] FIG. 3 illustrates an example of a flow diagram of a method, according to some exemplary embodiments; and

[0009] FIG. 4 illustrates a set of apparatuses, according to various exemplary embodiments.

DETAILED DESCRIPTION

[0010] It will be readily understood that the components of certain example embodiments, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. The following is a detailed description of some example embodiments of systems, methods, apparatuses, and non-transitory computer program products for group-based beam reporting (GBBR) for multi-receiver (Rx) operation based on reference signal received power (RSRP) differences. Although the devices discussed below and shown in the figures refer to 5G/6G or Next Generation NodeB (gNB) devices and user equipment (UE) devices, this disclosure is not limited to only gNBs and UEs.

[0011] It may be readily understood that the components of certain example embodiments, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Different reference designations from multiple figures may be used out of sequence in the description, to refer to a same element to illustrate their features or functions. If desired, the different functions or procedures discussed herein may be performed in a different order and/or concurrently with each other. Furthermore, if desired, one or more of the described functions or procedures may be optional or may be combined. As such, the following description should be considered as illustrative of the principles and teachings of certain example embodiments, and not in limitation thereof.

[0012] 3GPP technologies may define minimum specifications for UEs for multi-Rx chain downlink (DL) reception. A UE may have one or multiple antenna panels. In frequency range 2 (FR2), 4-layer multiple input multiple output (MIMO) reception requires beam reception from at least two directions. In FR2, carrier aggregation (CA) with independent beam management (IBM) and/or common beam management (CBM) may be supported with simultaneous DL reception on different component carriers from co-located and/or non-co-located transmission reception points (TRPs).

[0013] The UE may support IBM by performing DL simultaneous reception on different antenna panels using separate beams on different component carriers. A UE with IBM capabilities may have increased UE baseband and RF capabilities, such as multiple baseband chains and/or support of multiple antenna panels. Simultaneous reception may require support of simultaneous multi-panel operation(s) with multiple independent Rx beams and/or chains at the UE side. Multi-beam simultaneous reception and multiple Rx chains may provide performance improvements in FR2 by improving demodulation performance (4-layer DL MIMO), radio resource management (RRM) performance and/or radio frequency (RF) spherical coverage. Simultaneous reception may include situations in which dual data, dual reference signals (RF), and/or data and an RS are simultaneously

received.

[0014] Single transmission configuration indicator (TCI) reception on different beams has been supported by RAN 3GPP specifications using a Type I codebook, which may be achieved at the UE with either a single antenna panel or multiple antenna panels. Alternatively, dual TCI operation may be combined with a multiple TRP (mTRP) framework in, for example, a situation in which the base station is deployed as a single TRP.

[0015] GBBR may have a signaling framework provided for multi-TRP operation to enable UE measurement reporting of highest quality DL transmission (Tx) beam pairs and/or groups from a set of configured measurement resources. The set of configured measurement resources may be, for example, a synchronization signal block (SSB) or a channel state information reference signal (CSI-RS). In multi-TRP operations, a serving cell may schedule the UE from two TRPs, which may provide improved coverage, reliability and/or data rates for a physical downlink shared channel (PDSCH), a physical downlink control channel (PDCCH), a physical uplink shared channel (PUSCH), and a physical uplink control channel (PUCCH).

[0016] During GBBR, the UE may perform simultaneous DL data reception from different quasi co-location Type D (QCL-D) sources, as may be defined by 3GPP specifications. The UE may be configured with a radio resource control (RRC) configuration parameter (e.g., groupBasedBeamReporting) set as enabled. The UE may be configured with at least two different sets of Layer 1 (L1) measurement resources that are associated with two different active TRPs. The UE may measure the L1-RSRP from the RSs related to the at least two sets, constituting Tx beam pairs and/or groups. The UE may report the measurements in a single reporting instance, which may include one or more different groups of RSs, where the reporting of a group indicates that the UE may receive simultaneously from TCI states related to the group.

[0017] As an example, 3GPP specifications may define that an information element (e.g., CSI-ReportConfig) includes a parameter (e.g., groupBasedBeamReporting), and a parameter (e.g., groupBasedBeamReporting-r17) which may enable GBBR for mTRP. The parameter reportQuantity may be used for configuring contents of CSI-ReportConfig. For example, a parameter cri-RSRP may be used when the Channel Measurement Resource (CMR) sets have CSI-RS resources configured. As another example, a parameter ssb-Index-RSRP may be used when the CMR sets have SSB resources configured. The parameter nrofReportedGroups-r17 may be used to configure the number of Tx beam pairs and/or groups to be reported.

[0018] FIG. 1 illustrates an example of a network configuration for GBBR, according to various exemplary embodiments. This exemplary network configuration may include a UE **101**, a first TRP (TRP1) **102**, and a second TRP (TRP2) **103**. In this GBBR example, the network may transmit a first RS set (RS.sub.1, . . . , RS.sub.M1) from TRP1 **102** using different Tx beams (a.sub.1, . . . , a.sub.M1). The UE **101** may measure the first RS set (RS.sub.1, . . . , RS.sub.M1) using different Rx beams (c.sub.1, . . . , c.sub.N) and may determine a first Tx/Rx beam pair (a.sub.i, c.sub.j) for transmissions from TRP1 **102**, such as determining a first Tx/Rx pair which results in the largest L1-RSRP (RSRP.sub.ij). Similarly, the network may transmit a second RS set (RS'.sub.1, . . . , RS'.sub.M2) from TRP2 **103** using different Tx beams (b.sub.1, . . . , b.sub.M2). The UE **101** may measure the second RS set (RS'.sub.1, . . . , RS'.sub.M2) using different Rx beams (c.sub.1, . . . , c.sub.N) and may determine a second Tx/Rx beam pair (b.sub.k, c.sub.l) for transmissions from TRP2 **103**, such as determining a second Tx/Rx beam pair which results in the largest L1-RSRP (RSRP.sub.kl). The UE **101** may then report an RS group {RS.sub.i, RS'.sub.k} to the network, corresponding to the group of Tx beams {a.sub.i, b.sub.k}.

[0019] As the beam selection may be performed independently for each TRP, any potential cross-interference (RSRP.sub.kj, RSRP.sub.il) among the simultaneous DL transmissions may not be taken into account. In conventional GBBR operations, using the example configuration shown in FIG. 1, the Rx beams (c.sub.j, c.sub.l) may overlap significantly in space and may result in significant cross-interference between the DL transmissions using Tx/Rx beam pairs (a.sub.i,

c.sub.j) and (b.sub.k, c.sub.l).

[0020] Various example embodiments may provide technological advantages to account for RS cross-interference by evaluating whether a difference between the RSRP level measurements of multiple reference signals exceeds or does not exceed one or more thresholds. For example, for both Tx/Rx beam pairs (a.sub.i, c.sub.j) and (b.sub.k, c.sub.l), the received power of a desired RS (RSRP.sub.ij and RSRP.sub.kl, respectively) may need to be sufficiently stronger than the received power of an undesired RS (RSRP.sub.kj and RSRP.sub.il, respectively).

[0021] The RS cross-interference may be accounted for in the determination of whether a difference between the RSRP level measurements of multiple reference signals exceeds or does not exceed one or more thresholds using evaluations of each RS relative to each other. For example, the determination may be based on the following Equation (1) and Equation (2):

$$\text{RSRP.sub.ij} - \text{RSRP.sub.kj} > \text{threshold} \quad (1)$$

$$\text{RSRP.sub.kl} - \text{RSRP.sub.il} > \text{threshold} \quad (2)$$

[0022] Various example embodiments may provide one or more procedures to enhance a GBBR framework by taking into account potential cross-interference among at least two simultaneous DL transmissions to be received by a multi-Rx UE. For example, the potential cross-interference may be as a result of significant spatial overlap among the respective receive beams (c.sub.1, c.sub.2), a presence of high-gain side lobes, and/or the like. Certain exemplary embodiments may provide that the UE may determine at least two RSRP differences (A.sub.1, A.sub.2), one for each receive beam (c.sub.1, c.sub.2), by measuring reference signals (RS) (RS.sub.1, RS.sub.2) transmitted by the network with different transmit beams and/or from different locations (e.g., TRPs). The UE may then either report the RSRP differences (Δ .sub.1, Δ .sub.2) for consideration by the network or use the RSRP differences (Δ .sub.1, Δ .sub.2) to determine whether to exclude a corresponding candidate RS group ({RS.sub.1, RS.sub.2}) from a GBBR.

[0023] Some exemplary embodiments may provide one or more procedures for enhancing the GBBR framework to enable a better selection and reporting the strongest DL beam pair or group. Various exemplary embodiments may improve the quality of simultaneous receptions for a multi-Rx UE by taking into account potential cross-interference among the corresponding simultaneous DL transmissions, such as, for example, due to spatial overlap among the respective Rx beams at the UE. Using Layer-1 Signal-to-Interference-plus-Noise Ratio (L1-SINR) measurements may increase UE complexity and signaling overhead and may not accurately reflect the actual cross-interference potential between the simultaneous DL transmissions for the multi-RX UE. For example, a relatively low L1-SINR measurement may be caused by signals being transmitted in neighboring cells at the time of measurement.

[0024] Various exemplary embodiments may advantageously improve upon the GBBR reporting based on L1-RSRP measurements by ensuring that, for both Tx/Rx beam pairs (a.sub.i, c.sub.j) and (b.sub.k, c.sub.l), the received power of a desired RS (RSRP.sub.ij and RSRP.sub.kl, respectively) is sufficiently stronger than the received power of an undesired RS (RSRP.sub.kj and RSRP.sub.il, respectively). The L1-RSRP measurements may be already available at the UE, such as, for example, the L1-RSRP measurements may have been performed for the purpose of beam alignment with the network, thus dedicated measurements for cross-interference based GBBR may not be necessary. This may increase resource efficiency and save energy.

[0025] Some exemplary embodiments may provide that a UE may have at least two beams (c.sub.1, c.sub.2) available for simultaneous reception, such as, for example, one on each side of a handheld mobile UE. The UE may use a first beam (c.sub.1) to receive from a first TRP (TRP1) and a second beam (c.sub.2) to receive from a second TRP (TRP2). For a candidate RS group {RS.sub.i, RS.sub.2}, the UE may determine a first difference (Δ .sub.1=RSRP.sub.11−RSRP.sub.21) based on measurements on RS.sub.1 and RS.sub.2 using the

first beam (c.sub.1). The UE may also determine a second difference ($\Delta_{\text{sub.2}} = \text{RSRP}_{\text{sub.22}} - \text{RSRP}_{\text{sub.12}}$) based on measurements on RS.sub.i and RS.sub.2 using the second beam (c.sub.2). The measurements on RS.sub.1 and RS.sub.2 using a particular beam (c.sub.1, c.sub.2) may or may not be performed simultaneously, e.g., may be performed in parallel (in case RS.sub.1 and RS.sub.2 are transmitted simultaneously), or may be performed sequentially (in case RS.sub.1 and RS.sub.2 are transmitted at different times).

[0026] Certain exemplary embodiments may provide that the UE may have a number N of beams (c.sub.1, . . . , c.sub.N) available, from which any Rx beam pair (c.sub.j, c.sub.l) may be selected and/or used for simultaneous reception. Some exemplary embodiments may provide one or multiple Rx beam pairs which may be restricted to, for example, Rx beams on different panels which may be used simultaneously. For example, a first plurality or subset (S.sub.1) of candidate beams for receiving a first transmission may comprise Rx beams relating to a first antenna panel and a second plurality or subset (S.sub.2) of candidate beams for receiving a second transmission may comprise Rx beams relating to a second antenna panel. Various exemplary embodiments may also, or alternatively, provide that the one or multiple Rx beam pairs are not restricted, and the first and second pluralities or subsets (S.sub.1, S.sub.2) may be equal to a complete set (S) of all available beams at the UE (i.e., $S_{\text{sub.1}} = S_{\text{sub.2}} = S$).

[0027] A first difference ($\Delta_{\text{sub.1}}$) may be determined as a largest ($\Delta_{\text{sub.1}} = \max_{\text{sub.j}} \Delta_{\text{sub.j}}$) among the respective first differences ($\Delta_{\text{sub.j}} = \text{RSRP}_{\text{sub.ij}} - \text{RSRP}_{\text{sub.2j}}$) corresponding to the candidate Rx beam(s) (c.sub.j) from the first subset (S.sub.1). The UE may search for a first Rx beam (c.sub.j) which maximizes a received signal strength from RS.sub.1 (i.e., $\text{RSRP}_{\text{sub.ij}}$) while minimizing the received signal strength from RS.sub.2 (i.e., $\text{RSRP}_{\text{sub.2j}}$).

[0028] Further, a second difference ($\Delta_{\text{sub.2}}$) may be determined as the largest ($\Delta_{\text{sub.2}} = \max_{\text{sub.l}} \Delta_{\text{sub.1}}$) among the respective second differences ($\Delta_{\text{sub.1}} = \text{RSRP}_{\text{sub.2l}} - \text{RSRP}_{\text{sub.1l}}$) corresponding to the candidate Rx beams (c.sub.1) from the second subset (S.sub.2). The UE may search for a second Rx beam (c.sub.1) which maximizes the received signal strength from RS.sub.2 (i.e., $\text{RSRP}_{\text{sub.2l}}$) while minimizing the received signal strength from RS.sub.1 (i.e., $\text{RSRP}_{\text{sub.1l}}$).

[0029] Various exemplary embodiments may provide one or more thresholds (Th.sub.1, Th.sub.2) which may be configured (e.g., by the network) or pre-defined. For each candidate RS group {RS.sub.1, RS.sub.2}, the UE may compare the first and second differences ($\Delta_{\text{sub.1}}$, $\Delta_{\text{sub.2}}$) with the one or more threshold(s) (Th.sub.1, Th.sub.2) to determine whether the candidate RS group {RS.sub.1, RS.sub.2} may be included in or excluded from a GBBR. For example, as a result, the network may prevent the UE from reporting any candidate RS group {RS.sub.1, RS.sub.2} for which the first difference does not exceed a first threshold ($\Delta_{\text{sub.1}} < \text{Th}_{\text{sub.1}}$) and/or the second difference does not exceed a second threshold ($\Delta_{\text{sub.2}} < \text{Th}_{\text{sub.2}}$). This failure to exceed the first and/or second threshold may indicate to the network that the UE will not be able to sufficiently separate, via any pair of candidate Rx beams, the DL signals to be transmitted via the Tx beams associated with the candidate RS group {RS.sub.1, RS.sub.2}. After excluding the one or more candidate RS groups, the UE may use other and/or additional criteria to select which candidate RS groups to report, such as RSRP, channel rank, and/or the like.

[0030] The first and second thresholds (Th.sub.1, Th.sub.2) may depend on a modulation and coding scheme (MCS.sub.1, MCS.sub.2) to be used for the DL transmissions. For example, certain exemplary embodiments may provide a situation in which the network expects to use a robust MCS and the one or more thresholds may be configured to a lower value. Some exemplary embodiments may provide a situation in which the network may expect to use a relatively less robust MCS and the one or more thresholds may be configured to a relatively higher value.

[0031] The one or more thresholds may be based on a strength of the received power of the desired RS. For example, the first threshold (Th.sub.1) may be configured to a relatively lower value (i.e., more cross-interference from RS.sub.2 may be allowed) in case the received power of RS.sub.1 is strong, and/or the second threshold (Th.sub.2) may be configured to a higher value (i.e., less cross-

interference from RS.sub.1 may be allowed) in case the received power of RS.sub.2 is weak. The first and second thresholds (Th.sub.1, Th.sub.2) may be equivalent and/or a single threshold. [0032] Various exemplary embodiments may provide that the network may not impose restrictions on GBBR, and the network may configure the UE to report the first and second differences (Δ .sub.1, Δ .sub.2). The GBBR and the first and second differences (Δ .sub.1, Δ .sub.2) may be used by a DL scheduler, for example, when determining a suitable MCS for the simultaneous DL transmissions. A relatively smaller difference (Δ .sub.1 or Δ .sub.2) may cause the DL scheduler to select a more robust MCS for the corresponding DL transmission. A larger difference (Δ .sub.1 or Δ .sub.2) may cause the DL scheduler to select a less robust MCS, which may result in maximizing spectral efficiency.

[0033] In determining which candidate RS group(s) to report, certain exemplary embodiments may provide that the UE may evaluate and rank the candidate RS group(s) according to, for example, certain criteria and/or a ranking algorithm. The criteria and/or ranking algorithm may include a potential cross-interference. For example, the UE may assign a quality metric (ρ .sub.12) to a candidate RS group {RS.sub.1, RS.sub.2} based on the first and second differences (Δ .sub.1, Δ .sub.2). The quality metric (ρ .sub.12) may be based on the smaller one between the first difference (Δ .sub.1) and the second difference (Δ .sub.2) (i.e., $\min(\Delta$.sub.1, Δ .sub.2)). For example, a relatively higher quality metric (ρ .sub.12) may be assigned in case $\min(\Delta$.sub.1, Δ .sub.2) is large, while a relatively lower quality metric (ρ .sub.12) may be assigned in case $\min(\Delta$.sub.1, Δ .sub.2) is small. A relatively larger value of $\min(\Delta$.sub.1, Δ .sub.2) may mean that the UE has an improved capability to separate, via at least one pair of Rx beams, the DL signals to be transmitted via Tx beams associated with the candidate RS group {RS.sub.1, RS.sub.2}.

[0034] Once the quality metric (ρ .sub.12) has been determined for each candidate RS group {RS.sub.i, RS.sub.k}, the UE may include, in the GBBR, one or more candidate RS groups {RS.sub.i, RS.sub.k} with the highest quality metric (ρ .sub.ik), up to a configured number (e.g., $\text{nrOfReportedGroups-r17}$) of candidate RS groups to be reported. In some exemplary embodiments, a minimum quality metric (μ .sub.min) may be configured or pre-defined to prevent the UE from reporting a candidate RS group with a corresponding quality metric below a minimum.

[0035] Certain exemplary embodiments may provide that the UE may have a number N of beams (c.sub.1, . . . , c.sub.N) available, from which any subset of K Rx beams, or at least some of the possible subsets of K RX beams, may be used for simultaneous reception of K concurrent transmissions. For example, the UE may use each of the K Rx beams (c.sub.n, $n=1, \dots, K$) to receive from a different TRP out of K TRPs. For each candidate RS (RS.sub.n, $n=1, \dots, K$) in a given candidate RS group {RS.sub.1, RS.sub.2, . . . , RS.sub.K}, the UE may determine a corresponding difference (Δ .sub.n= $\text{RSRP.sub.mn}-\text{sum.sub.m}\neq\text{n RSRP.sub.mn}$) between an RSRP (RSRP.sub.mn) associated with the candidate RS (RS.sub.n) and a sum of all RSRPs ($\text{sum.sub.m}\neq\text{n RSRP.sub.mn}$) associated with all other candidate RSs (RS.sub.m, $m\neq n$). The RSRP measurements on all candidate RSs (RS.sub.1, RS.sub.2, . . . , RS.sub.K) may be performed using a (e.g., same) corresponding beam (c.sub.n) of the UE.

[0036] FIG. 2 illustrates a signal diagram for one or more procedures, according to various example embodiments. The one or more procedures may be performed by a configuration of, for example, a network entity or base station (e.g., gNB) **201**, a first TRP **202**, a second TRP **203**, and a UE **204**. The one or more procedures may provide, at **210**, the gNB **201** may provide one or more channel measurement resource (CMR) configurations to the UE **204**. At **220**, the one or more procedures may provide that the UE **204** may determine a candidate RS group {RS.sub.1, RS.sub.2} for a GBBR. At **230**, the TRP **202** may transmit a first RS.sub.1 to the UE **204**, which is received by the UE **204**, and at **240**, the TRP **203** may transmit a second RS.sub.2 to the UE **204**, which is received by the UE **204**.

[0037] Certain exemplary embodiments may provide, at **250**, the UE **204** may determine a first difference (Δ .sub.1= $\text{RSRP.sub.11}-\text{RSRP.sub.21}$) based on a first Rx beam c.sub.1, and at **260**, the

UE **204** may determine a second difference ($\Delta_{\text{sub.2}} = \text{RSRP}_{\text{sub.22}} - \text{RSRP}_{\text{sub.12}}$) based on a second Rx beam c.sub.2. At **270**, the UE **204** may determine whether to include or exclude a candidate RS group {RS.sub.1, RS.sub.2} for a GBBR based on the first and second differences ($\Delta_{\text{sub.1}}$ and $\Delta_{\text{sub.2}}$). At **280**, the UE **204** may transmit a GBBR to the gNB **201**, which may include the first and second differences ($\Delta_{\text{sub.1}}$ and $\Delta_{\text{sub.2}}$).

[0038] FIG. **3** illustrates an example flow diagram of a method, according to certain exemplary embodiments. In an example embodiment, the method of FIG. **3** may be performed by a network element, or a group of multiple network elements in a 3GPP system, such as LTE or 5G-NR. For instance, in an example embodiment, the method of FIG. **3** may be performed by a UE similar to apparatus **410** illustrated in FIG. **4**.

[0039] According to various exemplary embodiments, the method of FIG. **3** may include, at **310**, determining a candidate reference signal group to be provided to at least one other apparatus. The candidate reference signal group may include at least a first reference signal and a second reference signal. At **320**, the method may include determining, based on a first receive beam of the apparatus, a first difference between a first reference signal received power of the first reference signal and a second reference signal received power of the second reference signal. At **330**, the method may further include determining, based on a second receive beam of the apparatus, a second difference between a third reference signal received power of the second reference signal and a fourth reference signal received power of the first reference signal. At **340**, the method may include performing at least one of excluding the candidate reference signal group from a group-based beam report based on the first difference and the second difference, or transmitting, to the at least one other apparatus, information indicative of the first difference and the second difference.

[0040] Certain exemplary embodiments may provide that the method includes transmitting, to the at least one other apparatus, the group-based beam report comprising the candidate reference signal group. The method may also include determining, for each candidate receive beam from a first subset of candidate receive beams for receiving a first transmission, a respective first difference between a respective first reference signal received power of the first reference signal and a respective second reference signal received power of the second reference signal, and determining the first difference as the largest among the respective first differences corresponding to the candidate receive beams from the first subset. The method may include determining, for each candidate receive beam from a second subset of candidate receive beams for receiving a second transmission, a respective second difference between a respective third reference signal received power of the second reference signal and a respective fourth reference signal received power of the first reference signal, and determining the second difference as the largest among the respective second differences corresponding to the candidate receive beams from the second subset.

[0041] Some exemplary embodiments may provide that the first and second subset of candidate receive beams may be equal to a set of receive beams available at the apparatus. The method may also include determining whether the first difference and the second difference exceed at least one threshold, and perform at least one of, when the first difference and the second difference both exceed the at least one threshold, determining to further consider the candidate reference signal group for inclusion in the group-based beam report, or when at least one of the first difference or the second difference does not exceed the at least one threshold, determining to exclude the candidate reference signal group from the group-based beam report.

[0042] Various exemplary embodiments may provide that the at least one threshold is configured by a network entity. The at least one threshold may be based on a modulation and coding scheme. The at least one threshold may include a first threshold for the first difference and a second threshold for the second difference. The first threshold may be based on the first reference signal received power of the first reference signal and the second threshold may be based on the third reference signal received power of the second reference signal. The method may further include determining a quality metric for the candidate reference signal group based on at least the first

difference and the second difference. The method may also include determining the quality metric for the candidate reference signal group based on a smaller one among the first difference and the second difference.

[0043] Certain exemplary embodiments may provide that the method may include determining a plurality of candidate reference signal groups, which includes the candidate reference signal group. Each candidate reference signal group of the plurality of candidate reference signal groups may include at least corresponding first and second reference signals. The method may further include determining a corresponding quality metric for each candidate reference signal group of the plurality of candidate reference signal groups and transmitting the candidate reference signal groups with highest quality metric, up to a configured number of candidate reference signal groups to be reported, in the group-based beam report. The method may also include excluding, from the group-based beam report, one or more candidate reference signal groups of the plurality of candidate reference signal groups for which the corresponding quality metric is below a minimum quality metric threshold.

[0044] Various exemplary embodiments may provide that the method includes determining, for each candidate reference signal in one group of the plurality of candidate reference signal groups, a corresponding difference between a reference signal received power associated with the candidate reference signal and a sum of reference signal received powers associated with all other candidate reference signals in the one group of the plurality of candidate reference signal groups. The method may also include performing at least one of excluding the one group from the group-based beam report based on the corresponding differences for each candidate reference signal in the one group, or transmitting, to the at least one other apparatus, information indicative of the corresponding differences for each candidate reference signal in the one group.

[0045] FIG. 4 illustrates apparatuses **410** and **420** according to various example embodiments. In the various example embodiments, apparatus **410** may be an element in a network or associated with such a network, such as a UE. UE **302** may be an example of apparatus **410** according to various example embodiments as discussed above. It should be noted that one of ordinary skill in the art would understand that apparatus **410** may include components or features not shown in FIG. 4. Further, the apparatus **420** may be an element in a network or associated with such a network, such as a PCF. PCF **705** may be examples of apparatus **420** according to various example embodiments as discussed above. It should be noted that one of ordinary skill in the art would understand that apparatus **420** may include components or features not shown in FIG. 4.

[0046] According to various example embodiments, the apparatuses **410** and/or **420** may include one or more processors, one or more computer-readable storage medium (for example, memory, storage, or the like), one or more radio access components (for example, a modem, a transceiver, or the like), and/or a user interface. In some example embodiments, apparatuses **410** and/or **420** may be configured to operate using one or more radio access technologies, such as GSM, LTE, LTE-A, NR, 5G, WLAN, WiFi, NB-IoT, Bluetooth, NFC, MulteFire, and/or any other radio access technologies.

[0047] As illustrated in the example of FIG. 4, apparatuses **410** and/or **420** may include or be coupled to processors **412** and **422**, respectively, for processing information and executing instructions or operations. Processors **412** and **422** may be any type of general or specific purpose processor. In fact, processors **412** and **422** may include one or more of general-purpose computers, special purpose computers, microprocessors, digital signal processors (DSPs), field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), and processors based on a multi-core processor architecture, as examples. While a single processor **412** (**422**) for each of apparatuses **410** and/or **420** is shown in FIG. 4, multiple processors may be utilized according to other example embodiments. For example, it should be understood that, in certain example embodiments, apparatuses **410** and/or **420** may include two or more processors that may form a multiprocessor system (for example, in this case processors **412** and **422** may represent a

multiprocessor) that may support multiprocessing. According to certain example embodiments, the multiprocessor system may be tightly coupled or loosely coupled to, for example, form a computer cluster).

[0048] Processors **412** and **422** may perform functions associated with the operation of apparatuses **410** and/or **420**, respectively, including, as some examples, precoding of antenna gain/phase parameters, encoding and decoding of individual bits forming a communication message, formatting of information, and overall control of the apparatuses **410** and/or **420**, including processes illustrated in FIGS. 1-3.

[0049] Apparatuses **410** and/or **420** may further include or be coupled to memory **414** and/or **424** (internal or external), respectively, which may be coupled to processors **412** and **422**, respectively, for storing information and instructions that may be executed by processors **412** and **422**. Memory **414** (memory **424**) may be one or more memories and of any type suitable to the local application environment, and may be implemented using any suitable volatile or nonvolatile data storage technology such as a semiconductor-based memory device, a magnetic memory device and system, an optical memory device and system, fixed memory, and/or removable memory. For example, memory **414** (memory **424**) can be comprised of any combination of random access memory (RAM), read only memory (ROM), static storage such as a magnetic or optical disk, hard disk drive (HDD), or any other type of non-transitory machine or computer readable media. The instructions stored in memory **414** and memory **424** may include program instructions or computer program code that, when executed by processors **412** and **422**, enable the apparatuses **410** and/or **420** to perform tasks as described herein.

[0050] In certain example embodiments, apparatuses **410** and/or **420** may further include or be coupled to (internal or external) a drive or port that is configured to accept and read an external computer readable storage medium, such as an optical disc, USB drive, flash drive, or any other storage medium. For example, the external computer readable storage medium may store a computer program or software for execution by processors **412** and **422** and/or apparatuses **410** and/or **420** to perform any of the methods illustrated in FIGS. 1-3.

[0051] In some example embodiments, apparatuses **410** and/or **420** may also include or be coupled to one or more antennas **415** and **425**, respectively, for receiving a downlink signal and for transmitting via an uplink from apparatuses **410** and/or **420**. Apparatuses **410** and/or **420** may further include transceivers **416** and **426**, respectively, configured to transmit and receive information. The transceivers **416** and **426** may also include a radio interface (for example, a modem) respectively coupled to the antennas **415** and **425**. The radio interface may correspond to a plurality of radio access technologies including one or more of GSM, LTE, LTE-A, 5G, NR, WLAN, NB-IoT, Bluetooth, BT-LE, NFC, RFID, UWB, or the like. The radio interface may include other components, such as filters, converters (for example, digital-to-analog converters or the like), symbol demappers, signal shaping components, an Inverse Fast Fourier Transform (IFFT) module, or the like, to process symbols, such as OFDMA symbols, carried by a downlink or an uplink.

[0052] For instance, transceivers **416** and **426** may be respectively configured to modulate information on to a carrier waveform for transmission by the antenna(s) **415** and **425**, and demodulate information received via the antenna(s) **415** and **425** for further processing by other elements of apparatuses **410** and/or **420**. In other example embodiments, transceivers **416** and **426** may be capable of transmitting and receiving signals or data directly. Additionally or alternatively, in some example embodiments, apparatuses **410** and/or **420** may include an input and/or output device (I/O device). In certain example embodiments, apparatuses **410** and/or **420** may further include a user interface, such as a graphical user interface or touchscreen.

[0053] In certain example embodiments, memory **414** and memory **424** store software modules that provide functionality when executed by processors **412** and **422**, respectively. The modules may include, for example, an operating system that provides operating system functionality for

apparatuses **410** and/or **420**. The memory may also store one or more functional modules, such as an application or program, to provide additional functionality for apparatuses **410** and/or **420**. The components of apparatuses **410** and/or **420** may be implemented in hardware, or as any suitable combination of hardware and software. According to certain example embodiments, apparatus **410** may optionally be configured to communicate with apparatus **420** via a wireless or wired communications link **430** according to any radio access technology, such as NR.

[0054] According to certain example embodiments, processors **412** and **422**, and memory **414** and **424** may be included in or may form a part of processing circuitry or control circuitry. In addition, in some example embodiments, transceivers **416** and **426** may be included in or may form a part of transceiving circuitry.

[0055] For instance, in certain example embodiments, the apparatus **410** may be controlled by the memory **414** and the processor **412** to determine a candidate reference signal group to be provided to at least one other apparatus **420**. The candidate reference signal group may include at least a first reference signal and a second reference signal. The apparatus **410** may also be caused to determine, based on a first receive beam of the apparatus **410**, a first difference between a first reference signal received power of the first reference signal and a second reference signal received power of the second reference signal, and determine, based on a second receive beam of the apparatus **410**, a second difference between a third reference signal received power of the second reference signal and a fourth reference signal received power of the first reference signal. The apparatus **410** may further be caused to perform at least one of excluding the candidate reference signal group from a group-based beam report based on the first difference and the second difference, or transmitting, to the at least one other apparatus **420**, information indicative of the first difference and the second difference.

[0056] In some example embodiments, an apparatus (e.g., apparatus **410** and/or apparatus **420**) may include means for performing a method, a process, or any of the variants discussed herein. Examples of the means may include one or more processors, memory, controllers, transmitters, receivers, and/or computer program code for causing the performance of the operations.

[0057] Various example embodiments may be directed to an apparatus, such as apparatus **410**, that includes means for determining a candidate reference signal group to be provided to at least one other apparatus **420**. The candidate reference signal group may include at least a first reference signal and a second reference signal. The apparatus **410** may also include means for determining, based on a first receive beam of the apparatus **410**, a first difference between a first reference signal received power of the first reference signal and a second reference signal received power of the second reference signal, and means for determining, based on a second receive beam of the apparatus **410**, a second difference between a third reference signal received power of the second reference signal and a fourth reference signal received power of the first reference signal. The apparatus **410** may further include means for performing at least one of excluding the candidate reference signal group from a group-based beam report based on the first difference and the second difference, or transmitting, to the at least one other apparatus **420**, information indicative of the first difference and the second difference.

[0058] As used herein, the term “circuitry” may refer to hardware-only circuitry implementations (for example, analog and/or digital circuitry), combinations of hardware circuits and software, combinations of analog and/or digital hardware circuits with software/firmware, any portions of hardware processor(s) with software, including digital signal processors, that work together to cause an apparatus (for example, apparatus **410** and/or **420**) to perform various functions, and/or hardware circuit(s) and/or processor(s), or portions thereof, that use software for operation but where the software may not be present when it is not needed for operation. As a further example, as used herein, the term “circuitry” may also cover an implementation of merely a hardware circuit or processor or multiple processors, or portion of a hardware circuit or processor, and the accompanying software and/or firmware. The term circuitry may also cover, for example, a

baseband integrated circuit in a server, cellular network node or device, or other computing or network device.

[0059] A computer program product may include one or more computer-executable components which, when the program is run, are configured to carry out some example embodiments. The one or more computer-executable components may be at least one software code or portions of it. Modifications and configurations required for implementing functionality of certain example embodiments may be performed as routine(s), which may be implemented as added or updated software routine(s). Software routine(s) may be downloaded into the apparatus.

[0060] As an example, software or a computer program code or portions of it may be in a source code form, object code form, or in some intermediate form, and it may be stored in some sort of carrier, distribution medium, or computer readable medium, which may be any entity or device capable of carrying the program. Such carriers may include a record medium, computer memory, read-only memory, photoelectrical and/or electrical carrier signal, telecommunications signal, and software distribution package, for example. Depending on the processing power needed, the computer program may be executed in a single electronic digital computer or it may be distributed amongst a number of computers. The computer readable medium or computer readable storage medium may be a non-transitory medium.

[0061] In other example embodiments, the functionality may be performed by hardware or circuitry included in an apparatus (for example, apparatuses **410** and/or **420**), for example through the use of an application specific integrated circuit (ASIC), a programmable gate array (PGA), a field programmable gate array (FPGA), or any other combination of hardware and software. In yet another example embodiment, the functionality may be implemented as a signal, a non-tangible means that can be carried by an electromagnetic signal downloaded from the Internet or other network.

[0062] According to certain example embodiments, an apparatus, such as a node, device, or a corresponding component, may be configured as circuitry, a computer or a microprocessor, such as single-chip computer element, or as a chipset, including at least a memory for providing storage capacity used for arithmetic operation and an operation processor for executing the arithmetic operation.

[0063] The features, structures, or characteristics of example embodiments described throughout this specification may be combined in any suitable manner in one or more example embodiments. For example, the usage of the phrases “certain embodiments,” “an example embodiment,” “some embodiments,” or other similar language, throughout this specification refers to the fact that a particular feature, structure, or characteristic described in connection with an embodiment may be included in at least one embodiment. Thus, appearances of the phrases “in certain embodiments,” “an example embodiment,” “in some embodiments,” “in other embodiments,” or other similar language, throughout this specification do not necessarily refer to the same group of embodiments, and the described features, structures, or characteristics may be combined in any suitable manner in one or more example embodiments. Further, the terms “cell”, “node”, “gNB”, or other similar language throughout this specification may be used interchangeably.

[0064] As used herein, “at least one of the following: <a list of two or more elements>” and “at least one of <a list of two or more elements>” and similar wording, where the list of two or more elements are joined by “and” or “or,” mean at least any one of the elements, or at least any two or more of the elements, or at least all the elements.

[0065] One having ordinary skill in the art will readily understand that the disclosure as discussed above may be practiced with procedures in a different order, and/or with hardware elements in configurations which are different than those which are disclosed. Therefore, although the disclosure has been described based upon these example embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent, while remaining within the spirit and scope of example embodiments. Although the

above embodiments refer to 5G NR and LTE technology, the above embodiments may also apply to any other present or future 3GPP technology, such as LTE-advanced, and/or fourth generation (4G) and/or sixth (6G) technology.

Partial Glossary

[0066] 3GPP 3rd Generation Partnership Project [0067] 5G 5th Generation [0068] 6G 6.sup.th Generation [0069] BS Base Station [0070] CMR Channel Measurement Resource [0071] DL Downlink [0072] EMBB Enhanced Mobile Broadband [0073] FR2 Frequency Range 2 [0074] gNB 5G or Next Generation NodeB [0075] ID Identifier [0076] L1 Layer 1 [0077] LTE Long Term Evolution [0078] MCS Modulation and Coding Scheme [0079] NR New Radio [0080] QCL-D Quasi Co-Located Type D [0081] RS Reference Signal [0082] RSRP Reference Signal Received Power [0083] SINR Signal-to-Interference-plus-Noise Ratio [0084] TCI Transmission Configuration Indication [0085] TRP Transmission/Reception Point [0086] UE User Equipment [0087] UL Uplink

Claims

1. An apparatus comprising: at least one processor; and at least one memory storing instructions that, when executed by the at least one processor, cause the apparatus at least to: determine a candidate reference signal group to be provided to at least one other apparatus, wherein the candidate reference signal group comprises at least a first reference signal and a second reference signal; determine, based on a first receive beam of the apparatus, a first difference between a first reference signal received power of the first reference signal and a second reference signal received power of the second reference signal; determine, based on a second receive beam of the apparatus, a second difference between a third reference signal received power of the second reference signal and a fourth reference signal received power of the first reference signal; and perform at least one of: excluding the candidate reference signal group from a group-based beam report based on the first difference and the second difference, or transmitting, to the at least one other apparatus, information indicative of the first difference and the second difference.
2. The apparatus according to claim 1, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: transmit, to the at least one other apparatus, the group-based beam report comprising the candidate reference signal group.
3. The apparatus according to claim 1, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: determine, for each candidate receive beam from a first subset of candidate receive beams for receiving a first transmission, a respective first difference between a respective first reference signal received power of the first reference signal and a respective second reference signal received power of the second reference signal; and determine the first difference as the largest among the respective first differences corresponding to the candidate receive beams from the first subset.
4. The apparatus according to claim 3, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: determine, for each candidate receive beam from a second subset of candidate receive beams for receiving a second transmission, a respective second difference between a respective third reference signal received power of the second reference signal and a respective fourth reference signal received power of the first reference signal; and determine the second difference as the largest among the respective second differences corresponding to the candidate receive beams from the second subset.
5. The apparatus according to claim 4, wherein the first and second subset of candidate receive beams are equal to a set of receive beams available at the apparatus.
6. The apparatus according to claim 1, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: determine whether the first difference and the second difference exceed at least one threshold; and perform at least one of: when the first difference and

the second difference both exceed the at least one threshold, determining to further consider the candidate reference signal group for inclusion in the group-based beam report; or when at least one of the first difference or the second difference does not exceed the at least one threshold, determining to exclude the candidate reference signal group from the group-based beam report.

7. The apparatus according to claim 6, wherein the at least one threshold is configured by a network entity.

8. The apparatus according to claim 7, wherein the at least one threshold is based on a modulation and coding scheme.

9. The apparatus according to claim 6, wherein the at least one threshold comprises a first threshold for the first difference and a second threshold for the second difference.

10. The apparatus according to claim 9, wherein the first threshold is based on the first reference signal received power of the first reference signal and the second threshold is based on the third reference signal received power of the second reference signal.

11. The apparatus according to claim 1, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: determine a quality metric for the candidate reference signal group based on at least the first difference and the second difference.

12. The apparatus according to claim 11, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: determine the quality metric for the candidate reference signal group based on a smaller one among the first difference and the second difference.

13. The apparatus according to claim 1, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: determine a plurality of candidate reference signal groups, which includes the candidate reference signal group, wherein each candidate reference signal group of the plurality of candidate reference signal groups comprises at least corresponding first and second reference signals; determine a corresponding quality metric for each candidate reference signal group of the plurality of candidate reference signal groups; and transmit the candidate reference signal groups with highest quality metric, up to a configured number of candidate reference signal groups to be reported, in the group-based beam report.

14. The apparatus according to claim 13, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: exclude, from the group-based beam report, one or more candidate reference signal groups of the plurality of candidate reference signal groups for which the corresponding quality metric is below a minimum quality metric threshold.

15. The apparatus according to claim 13, wherein the instructions, when executed by the at least one processor, further cause the apparatus to: determine, for each candidate reference signal in one group of the plurality of candidate reference signal groups, a corresponding difference between a reference signal received power associated with the candidate reference signal and a sum of reference signal received powers associated with all other candidate reference signals in the one group of the plurality of candidate reference signal groups; and perform at least one of: excluding the one group from the group-based beam report based on the corresponding differences for each candidate reference signal in the one group, or transmitting, to the at least one other apparatus, information indicative of the corresponding differences for each candidate reference signal in the one group.

16. A method, comprising: determining, by a first apparatus, a candidate reference signal group to be provided to at least one second apparatus, wherein the candidate reference signal group comprises at least a first reference signal and a second reference signal; determining, by the first apparatus and based on a first receive beam of the first apparatus, a first difference between a first reference signal received power of the first reference signal and a second reference signal received power of the second reference signal; determining, by the first apparatus and based on a second receive beam of the first apparatus, a second difference between a third reference signal received power of the second reference signal and a fourth reference signal received power of the first reference signal; and performing at least one of: excluding the candidate reference signal group

from the group-based beam report based on the first difference and the second difference, or transmitting, to the at least one second apparatus, information indicative of the first difference and the second difference.
